JIYUN JANG

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RESEARCH INTEREST

My long-term research goal is to develop general-purpose robotic agents that can live and interact with humans in unstructured real-world environments. To this end, my research focuses on improving the generalizability of manipulation policy models (e.g. VLA, WAM), enabling robots to reliably execute diverse tasks under distribution shifts in objects, viewpoints, and environments. In particular, I explore methods for improving action-space grounding and robustness under distribution shifts. Through this line of research, I aim to contribute to scalable learning frameworks for reliable human-compatible robots.

Keywords: VLA, WAM, Robotic Reinforcement Learning, Human-Robot Interaction(HRI)

EDUCATION

Korea University *Mar. 2025 – Feb. 2027 (Expected)*
M.S. in Computer Science and Engineering
Research Advisors: Prof. Jinkyu Kim and Prof. Jungbeom Lee
Research Collaborations: Prof. Sungjoon Choi (Korea University), Jensen Gao (Stanford University)
Current GPA: 4.42/4.50

Stanford University *Sep. 2026 – Dec. 2026 (Scheduled)*
Visiting Student Researcher in ILIAD Lab
Research Advisors: Prof. Dorsa Sadigh and Jensen Gao

Korea University *Mar. 2021 – Feb. 2025*
B.E. in Computer Science and Engineering
GPA: 4.12/4.50

PUBLICATIONS

[P2] AxisGuide: Grounding Robot Action Coordinate System in RGB Observations for Robust Visuomotor Manipulation
Jiyun Jang, Yujin Sung, Woosung Joung, Daewon Chae, Sangwon Lee, Sohwi Kim, Jinkyu Kim, Jungbeom Lee
Robotics: Science and Systems (RSS), 2026

[P1] Finetuning Pre-trained Model with Limited Data In 3D Object Detection by Bridging Domain Gaps
Jiyun Jang, Mincheol Chang, Jongwon Park, Jinkyu Kim
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2024

PROJECTS

VR Teleoperation System for UR5e Manipulation

- Developed a teleoperation interface using Meta Quest 3 for intuitive control of a UR5e robot via VR controller pose tracking.
- Implemented a UR5e simulator and established a teleoperation pipeline between VR and simulation for safe testing and development.
- Extended the system to enable teleoperation of the real UR5e robot, bridging VR control with real-world manipulation.
- Built a unified framework connecting VR, simulation, and real robot platforms for data collection and manipulation research.

Interactive Target-conditioned Pick-and-Place with VLA Models

- Tackled the challenge of specifying user-desired placement locations in pick-and-place tasks, where conventional VLA policies lack an interface to receive explicit spatial targets.

- Developed an interactive manipulation system on a UR5e robot that allows users to specify the desired placement position by clicking on the front-view image.
- Introduced a visual target marker by overlaying the clicked pixel as a red marker on the image and providing it as an additional input to the VLA model.
- During training, detected the final object position from trajectories and incorporated it into demonstrations to enable the model to learn the correspondence between visual target markers and placement actions.

Deformable Object Manipulation with Robotic RL

- Tackled the challenge of deformable object manipulation, where realistic simulation is difficult, by building a reset-free real-world RL pipeline that eliminates the need for manual resets.
- Developed a vision-based reinforcement learning framework on a Franka Emika Panda robot.
- This project is granted by IITP and Samsung Electronics DS (Device Solutions).

Preference-based RL for Humanoid Motion Generation

- Tackled the challenge of designing rewards for human-like humanoid motions, where explicit reward specification is difficult, by adopting preference-based reinforcement learning.
- Developed a preference-based RL framework to generate novel, human-like basketball shooting motions for a humanoid robot.
- Implemented the framework based on "PreferenceTransformer" (*ICLR* 2023) and evaluated it in "Humanoid-Bench" (*RSS* 2024).

Domain Adaptation on LiDAR Configuration for 3D Object Detection

- Tackled severe performance degradation caused by domain shifts from varying LiDAR configurations (sensor position and orientation) by proposing a test-time domain adaptation framework.
- Developed a LiDAR-based 3D object detection adaptation method that improves robustness across different sensor setups.
- This project was conducted in collaboration with the HYUNDAI Motor Autonomous Driving Center.

Unsupervised 3D Human Detection

- Vision and AI Lab Challenge, 3D Human Detection in Solid-state LiDAR data
- *2nd place* by Heuristic Point Cloud Clustering Algorithm (2,000 $)

Virtual AI Assistant

- Application for Weather asking, Screen recording, GPT query by Voice Activity Detection
- TTS Finetuning & AI Application Optimization
- *Excellence award* in AIKU Winter Conference 2023

EXTRA-CURRICULAR ACTIVITIES

- Korea University Silicon Valley Program *Jun. 2024 - Jul. 2024*
- Korea University AI club (AIKU) member *Jan. 2023 - Dec. 2023*
- Student President of Korea University College of Informatics *Nov. 2021 - Nov. 2022*

SKILLS

Programming Languages
C, C++, Python (PyTorch, TensorFlow, JAX), C# (Unity)

Strengths
Leadership, Initiative, Self-Motivation

TEACHING EXPERIENCE

Teaching Assistant

COSE111 Linear Algebra — *Spring 2026*

COSE112 Probability and Statistics — *Fall 2025*